$$uv = \int udv + \int vdu$$

$$\therefore \int udv = uv - \int vdu$$

This method is very useful when the integrand is a product of two different types of functions or a function which is not directly integrable. The success of this method depends on the proper choice of u. So we can choose the function u by using the following guidelines.

(i) If the integrand contains only a function which is directly not integrable, then take this as u.

(ii) If the integrand contains both directly integrable and non integrable functions, then take non integrable function as u.

(iii) If the integrand contains both the functions are integrable and one of them is of the form x^n, n is a positive integer, then take this x^n as u.

(iv) for all other cases, the choice of u is ours (Or) we can also choose u as the function which comes first in the word **"I L A T E"** Where,

I stands for the inverse trigonometric function
L stands for the logarithmic function
A stands for the algebraic function
T stands for the trigonometric function
E stands for the exponential function and take the remaining part of the function and dx as dv.

(2) If u and v are functions of x, then
$\int udv = uv - u'v_1 + u''v_2 - u'''v_3 + \ldots$

where $u', u'', u''', \ldots$ are the successive derivatives of u and $v_1, v_2, v_3, \ldots$ are the repeated integrals of v.

Note

- The above mentioned formula is well known as Bernoulli's formula.
- Bernoulli's formula is applied when $u = x^n$ where n is a positive integer.

Example 2.25

Evaluate $\int xe^x\,dx$

Solution:

Take $u = x$	and $dv = e^x dx$
Differentiate	Integrate
$du = dx$	$v = e^x$

$$\begin{aligned}\int xe^x\,dx &= \int udv \\ &= uv - \int vdu \\ &= xe^x - \int e^x dx \\ &= xe^x - e^x + c \\ &= e^x(x-1) + c\end{aligned}$$

Example 2.26

Evaluate $\int x^3 e^x dx$

Solution:

Successive derivatives	Repeated integrals
	and $dv = e^x dx$
Take $u = x^3$	$v = e^x$
$u' = 3x^2$	$v_1 = e^x$
$u'' = 6x$	$v_2 = e^x$
$u''' = 6$	$v_3 = e^x$

$$\int x^3 e^x dx = \int udv$$

$$= uv - u'v_1 + u''v_2 - u'''v_3 + \ldots$$

$$= x^3e^x - 3x^2e^x + 6xe^x - 6e^x + c$$

$$= e^x\left(x^3 - 3x^2 + 6x - 6\right) + c$$

Example 2.27

Evaluate $\int x^3 \log x\,dx$

Solution:

Take $u = \log x$	and $dv = x^3 dx$
Differentiate	Integrate
$du = \frac{1}{x}dx$	$v = \frac{x^4}{4}$

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